

1. Evaluate the following expressions.

[2] (a) $10 - 2(-7 + (-2)^4)$

[2] (b) $\frac{5}{6} \div \frac{-6 + (-2^2)}{|-3|} + \frac{7}{2}$

2. Simplify the following polynomial expressions.

[2] (a) $\frac{3}{5}xy + 6xyz + \frac{7}{10}xy - 9xyz$

[3] (b) $(3x + 2)^2 - 4(x + 5)(x - 5)$

3. Solve for x in the following equations or show that there are no or infinitely many solutions. If any fractions occur in your answers, they should be fully simplified.

[3] (a) $2[x + 3(2x - 5)] + 1 = 8x - 1$

[3] (b) $(x + 4)(2x - 1) = 2x^2 + 12x - 4$

[3] (c) $\frac{4x + 6}{5} - \frac{4 - 3x}{2} = \frac{3}{2}$

4. Factor the following polynomials completely.

[2] (a) $x^2 - 15x + 50$

[2] (b) $25 - 81x^2$

[3] (c) $2x^5 - 32x$

5. Solve each of the following equations by **factoring**.

[3] (a) $8x^2 - 39x - 5 = 0$

[4] (b) $x^3 + 9x^2 - 4 = -2(9x + 2)$

6. Simplify the following expressions and present the results without negative exponents. You may assume that all variables are non-zero.

[1] (a) $-9x^{-2}$

[1] (b) $\frac{15x^3y^2}{20xy^5}$

[2] (c) $\left(\frac{20x^{-1}y}{60xyz^8}\right)^{-2}$

[2] (d) $x(y^2z^{-3})^5(-5xy^0z^{-5})^{-2}$

[3] 7. Solve the following equation using **the quadratic formula**. Simplify your answer(s) completely.

$$2x^2 + 10x + 5 = 0$$

8. Simplify each of the following expressions. You may assume that all variables represent positive numbers.

[3] (a) $x^5y^4\sqrt{75xy} + 7\sqrt{3x^{11}y^9}$

[3] (b) $\frac{\sqrt{24x^{12}y^5z^8}}{\sqrt{2x^6y^7z^3}}$

9. Rationalize the denominator of each expression and simplify completely.

[2] (a) $\frac{6}{\sqrt{3x}}$

[2] (b) $\frac{9}{2 - \sqrt{7}}$

10. Solve the following equations or show that there are no solutions.

[3] (a) $9 - \sqrt{21 - 3x} = 0$

[3] (b) $\sqrt{x + 7} - 1 = x$

11. You are given two points, $A = (3, -2)$ and $B = (2, 5)$.

[2] (a) Find the distance between A and B . Simplify your answer completely.

[1] (b) Find the midpoint of the line segment connecting A and B .

[2] 12. Draw the line $y = \frac{1}{3}x - 2$ on the given grid.

[3] 13. Find the x -intercept, y -intercept, and slope of the line $\mathcal{L} : 3x - 5y = 30$

[3] 14. Find an equation in slope-intercept form for the line $\mathcal{L}_1$ which passes through the point $(-4, 7)$ and is parallel to the line $\mathcal{L}_2 : y = -\frac{3}{2}x + 8$.

[2] 15. Determine whether the lines $\mathcal{L}_1 : y = 5x + \frac{1}{2}$ and $\mathcal{L}_2 : x + 5y = -15$ are parallel, perpendicular or neither.

[3] 16. Solve the following linear system by the method of **substitution**.

$$\begin{cases} 3x - y = 12 \\ -7x + 4y = -23 \end{cases}$$

[3] 17. Solve the following linear system by the method of **elimination**.

$$\begin{cases} 2x + 4y = 12 \\ 3x + 5y = 14 \end{cases}$$

[2] 18. Evaluate the following expression: $\ln e + \log_2 8 + \log_3 1 + \log_5 \left(\frac{1}{25}\right)$

19. Solve each equation for x .

[3] (a) $4^{7x+1} = 32^{3-2x}$

[3] (b) $\frac{3^{2x+6}}{3^{x+1}} = 5$

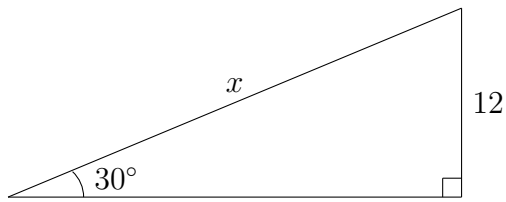
[3] 20. Let θ be an acute angle in a right triangle. Given that $\tan \theta = \frac{3}{4}$, find:

(a) $\cot \theta$

(b) $\cos \theta$

[1] 21. Evaluate and simplify: $(\sec(30^\circ))^2$

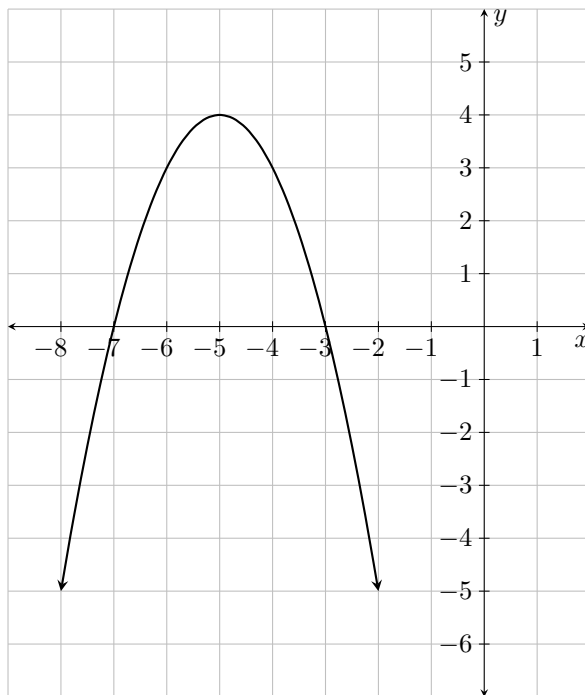
- [3] **22.** Use trigonometry to find the value of x in the triangle below. Simplify your answer as much as possible.
(Note that the illustration is not necessarily to scale.)



- [4] **23.** Given $f(x) = 2x^2$ and $g(x) = |3x|$, evaluate and simplify the following expressions.

- (a) $f(-5) + g(-2)$
(b) $f(a + h)$

- [5] **24.** Given the graph of the function $f(x)$, determine the characteristics below.



- (a) Domain
(b) Range
(c) x -intercept(s)
(d) Interval(s) over which $f(x)$ is positive
(e) Interval(s) over which $f(x)$ is increasing

Answers are on the next page

Answers:

1. (a) -8 (b) $\frac{13}{4}$

2. (a) $\frac{13}{10}xy - 3xyz$ (b) $5x^2 + 12x + 104$

3. (a) $x = \frac{14}{3}$ (b) $x = 0$ (c) $x = 1$

4. (a) $(x - 10)(x - 5)$
(b) $(5 + 9x)(5 - 9x)$
(c) $2x(x + 2)(x - 2)(x^2 + 4)$

5. (a) $x = 5, -\frac{1}{8}$ (b) $x = -6, -3, 0$

6. (a) $\frac{-9}{x^2}$ (b) $\frac{3x^2}{4y^3}$ (c) $9x^4z^{16}$ (d) $\frac{y^{10}}{25xz^5}$

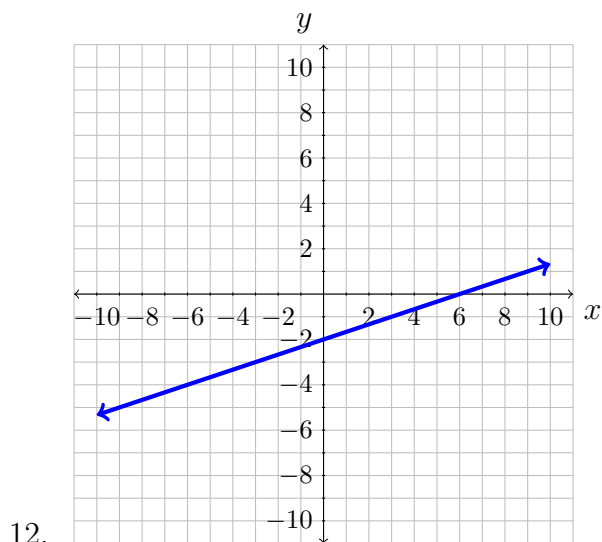
7. $x = \frac{-5 \pm \sqrt{15}}{2}$

8. a) $12x^5y^4\sqrt{3xy}$ b) $\frac{2x^3z^2\sqrt{3z}}{y}$

9. a) $\frac{2\sqrt{3x}}{x}$ b) $-6 - 3\sqrt{7}$

10. a) $x = -20$ b) $x = 2$

11. a) $5\sqrt{2}$ b) $(\frac{5}{2}, \frac{3}{2})$



13. x -int is $(10, 0)$, y -int is $(0, -6)$, slope is $\frac{3}{5}$.

14. $y = -\frac{3}{2}x + 1$

15. They are perpendicular (the slopes are 5 and $-\frac{1}{5}$).

16. $(5, 3)$

17. $(-2, 4)$

18. $1 + 3 + 0 + (-2) = 2$

19. a) $x = \frac{13}{24}$ b) $x = \log_3(5) - 5$

20. a) $\frac{4}{3}$ b) $\frac{4}{5}$

21. $\frac{4}{3}$

22. $x = 24$

23. a) 56 b) $2a^2 + 4ah + 2h^2$

24. a) $x \in (-\infty, \infty)$ b) $y \in (-\infty, 4]$
c) $(-7, 0)$ and $(-3, 0)$ d) $x \in (-7, -3)$
e) $x \in (-\infty, -5)$